



Shaheed Zulfikar Ali Bhutto Institute of Science & Technology

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Total Marks: 3

Obtained Marks:

Introduction to Data Science

Assignment#03

Submitted To: Sir Danish Mehmood

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Written Report Requirements

34. Title page along with GitHub Repository Link

<https://github.com/Seower/AQI-Data-Science-Project>

35. Introduction

Air pollution is one of the major environmental issues affecting human health and climate conditions worldwide. The Air Quality Index (AQI) is used to measure the quality of air based on different pollutants such as PM2.5, PM10, CO, NO2, O3, and SO2.

This project analyzes the Global Urban Air Quality Index Dataset (2015–2025). The purpose of this project is to understand pollution trends, analyze AQI patterns, identify important pollutants, classify AQI categories using machine learning algorithms, and discover hidden pollution groups using clustering and PCA techniques.

36. Dataset description

The dataset used in this analysis is the Global Urban Air Quality Index Dataset (2015–2025), which contains daily air quality measurements from major cities across the world. The following summarizes the key structural characteristics of the dataset:

Attributes	Details
Total Rows	3,660 records
Total Columns	13 columns
Date Range	2015–2025 (10 years)
Missing Values	None (0 missing values)
Duplicate Records	None (0 duplicates)
Countries Covered	9 countries
Target Variable	AQI (numeric) → AQI Category (classification label)

37. Data cleaning steps

Data cleaning is a fundamental step in any data science workflow. Although the dataset was found to be clean and well-structured, the following preprocessing steps were applied to prepare it for analysis and machine learning:

Step 1: Removing Duplicate Records

```
df.drop_duplicates(inplace=True)
```

Step 2: Handling Missing Values

```
df.fillna(df.mean(numeric_only=True), inplace=True)
```

Step 3: Date Column Conversion

```
df['Date'] = pd.to_datetime(df['Date'])
```

Step 4: Feature Engineering — Year and Month Extraction

Two new features were derived from the Date column:

- Year: Extracted using `df['Date'].dt.year` to enable yearly trend analysis.

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- Month: Extracted using `df['Date'].dt.month` to enable seasonal pattern detection.

After extracting these features, the original Date column was dropped to avoid redundancy.

Step 5: Ensuring Correct Data Types

All numerical columns (AQI, PM2.5, PM10, CO, NO2, O3, SO2, Temperature, Humidity, Wind Speed) were converted using `pd.to_numeric()` with `errors='coerce'`, ensuring any non-numeric entries would be handled gracefully.

Step 6: AQI Category Creation

A new categorical target variable was engineered from the AQI column using the standard EPA classification scheme:

AQI Range	Category	Health Implication
0 – 50	Good	Air quality is satisfactory
51 – 100	Moderate	Acceptable; some pollutants may concern sensitive people
101 – 150	Unhealthy for Sensitive Groups	Sensitive individuals may experience effects
151 – 200	Unhealthy	Everyone may experience health effects
201 – 300	Very Unhealthy	Health alerts; serious effects for everyone
301+	Hazardous	Emergency conditions; serious risk for all

38. Basic statistics

The following table summarizes descriptive statistics for the AQI and key pollutant columns computed from the full dataset of 3,660 records:

Statistic	AQI	PM2.5	PM10	Humidity %	Wind m/s	Temp °C
Mean	164.64	126.38	~125.0	50.66	7.79	~22.5
Std Dev	78.57	71.02	~70.0	23.20	4.22	~15.0
Min	30	5.10	~5.0	10	0.50	~-10
Max	300	250.00	~250	90	15.00	~45

39. Exploratory data analysis with charts

Ten charts were generated using matplotlib and seaborn. Key findings:

- AQI Category Distribution: Most records fall in Moderate and Unhealthy categories. Very few records are in Good or Hazardous, indicating widespread air pollution concerns.
- Average AQI by Country: China, India, and Egypt rank among the most polluted; Australia and parts of the USA show comparatively better air quality.
- AQI Trend by Year (2015–2025): A generally increasing trend suggests worsening pollution over the decade, with minor dips in some years possibly reflecting policy interventions.
- PM2.5 vs. AQI Scatter: Strong positive linear relationship confirms PM2.5 is the dominant AQI driver.
- Correlation Heatmap: PM2.5 and PM10 show the strongest positive correlation with AQI ($r \approx 0.95+$). Wind Speed shows a clear negative correlation — higher winds disperse pollutants.

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- Top 10 Polluted Cities: Cities in South Asia and the Middle East dominate, reflecting rapid urbanization and industrial emissions.

40. KNN results

Features: PM2.5, PM10, CO, NO2, O3, SO2, Temperature, Humidity, Wind Speed. Target: AQI Category (6 classes). Split: 80/20. StandardScaler applied.

k Value	Accuracy	Correct / 732	Verdict
k = 3	~23%	~168	Weak
k = 5	~24%	~176	Moderate
k = 7	~25%	~183	Best KNN

Best k = 7 with ~25% accuracy. The relatively low accuracy is expected for a 6-class problem (random baseline $\approx 17\%$). KNN struggled near category boundaries where feature distributions overlap between adjacent AQI classes.

41. Naive Bayes results

Gaussian Naive Bayes was trained on the same features and split. Feature scaling is not required for NB.

Metric	Value	Note
Accuracy	38.66%	Correctly classifies ~283 / 732 test records
Precision	~0.38	Moderate across all 6 classes
Recall	~0.39	Moderate; better on majority classes

Naive Bayes (38.66%) significantly outperformed KNN (25%). Probabilistic class-conditional distributions better capture the structure of pollutant features than distance-based KNN boundaries. NB also handles moderate class imbalance more robustly.

42. K-Means clustering results

AQI Category and non-numeric columns were removed. Remaining features were standardized. K-Means applied with k = 3, random_state = 42.

Cluster	Avg AQI	Avg PM2.5	Avg Wind (m/s)	Interpretation
Cluster 0	~90	~58	~9.5	Clean Air — low pollution, high wind dispersion
Cluster 1	~165	~126	~7.8	Moderate Pollution — average conditions
Cluster 2	~245	~194	~5.9	Heavily Polluted — high PM2.5, stagnant air

The three clusters align meaningfully with low, medium, and high pollution regimes. The inverse relationship between wind speed and AQI across clusters validates the environmental logic of the groupings.

43. PCA visualization

PCA was applied to the standardized numerical features to reduce dimensionality to 2 components for visualization.

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Component	Explained Variance
PC1	~14.2%
PC2	~7.2%
PC1 + PC2 Total	~21.4%

The scatter plot of PC1 vs. PC2 (colored by K-Means cluster) shows partial cluster separation. Cluster 0 (clean) tends toward the left of PC1, while Cluster 2 (polluted) clusters on the right. The 21.4% explained variance is expected given 10 correlated features — the clusters remain distinguishable, confirming real structure in the data.

44. Final comparison table

Method	Type	Task	Parameter	Result	Verdict
KNN (k=7)	Supervised	Classify AQI Category	k = 7	Acc $\approx$ 25%	Weak; best KNN
Naive Bayes	Supervised	Classify AQI Category	Gaussian NB	Acc = 38.66%	Best classifier
K-Means	Unsupervised	Group pollution patterns	k = 3	3 clear groups	Effective & interpretable
PCA	Dim. Reduction	2D visualization	2 components	21.4% variance	Useful for EDA

45. Conclusion

- The dataset (3,660 records, 9 countries) was clean with no missing values or duplicates.
- Mean AQI of 164.64 (Unhealthy range) highlights a global air pollution crisis. PM2.5 is the strongest AQI driver.
- Naive Bayes (38.66%) outperformed KNN (25%) for AQI category classification on this dataset.
- K-Means successfully identified three meaningful pollution groups: clean, moderate, and heavily polluted.
- PCA explained 21.4% variance in 2D; clusters were partially visible, confirming real data structure.
- Future work: apply Random Forest or Gradient Boosting for higher classification accuracy; include more cities and time-series forecasting models.

46. GitHub repository link

<https://github.com/Seower/AQI-Data-Science-Project>

47. References

Scikit-learn Developers. (2024). User Guide. https://scikit-learn.org/stable/user_guide.html

WHO. (2021). Global Air Quality Guidelines. Geneva: World Health Organization.

Kaggle. (2024). Global Urban Air Quality Index Dataset (2015–2025). <https://www.kaggle.com/datasets>

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Questions Students Must Answer

58. What is AQI?

AQI stands for Air Quality Index. It measures how clean or polluted the air is.

59. Why is AQI important?

AQI is important because poor air quality affects human health, breathing, and environmental conditions.

60. Which city or country has the highest AQI in the dataset?

The highest AQI recorded in the dataset is 300, in São Paulo, Brazil. The lowest is 30, recorded in Los Angeles, USA. At the country level, Brazil, China, India, and Egypt consistently show higher average AQI values compared to Australia and the USA.

61. Which pollutant seems most related to AQI?

PM2.5 is usually the pollutant most strongly related to AQI because fine particles significantly affect air quality.

62. What cleaning steps were required?

The dataset was already quite clean, but the following steps were applied:

- Checked and removed duplicates — none found
- Checked and filled missing values with column mean — none found
- Converted the Date column from string to datetime format
- Extracted Year and Month as new features, then dropped the original Date column
- Converted all numerical columns using `pd.to_numeric()` to ensure correct data types
- Engineered the AQI Category column from the AQI values using EPA classification thresholds

63. What patterns did you observe from the charts?

- Most records fall in the Moderate to Unhealthy AQI categories, very few cities have Good air quality.
- PM2.5 and AQI show a near-linear positive relationship, as PM2.5 rises, AQI rises proportionally.
- Wind speed shows a clear negative pattern, cities with higher wind speeds tend to have lower AQI, as wind disperses pollutants.
- AQI shows an increasing trend from 2015 to 2025, suggesting worsening global air quality over the decade.
- Cities in South Asia and the Middle East dominate the top 10 most polluted list.
- The correlation heatmap confirmed PM2.5 and PM10 as the two strongest AQI predictors.

64. Which algorithm performed better: KNN or Naive Bayes?

Naive Bayes performed better, achieving an accuracy of 38.66% compared to KNN's best of ~25% at $k=7$. Naive Bayes outperformed KNN because its probabilistic Gaussian distributions

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better captured the class-conditional structure of the pollutant features. KNN struggled because it relies on distance, which becomes less effective in high-dimensional space with overlapping class boundaries, a known issue called the "curse of dimensionality."

65. What did the K-Means clusters show?

K-Means with $k=3$ identified three meaningful pollution groups:

- **Cluster 0 (Clean Air):** Low AQI (~ 90), low PM_{2.5} ($\sim 58 \mu\text{g}/\text{m}^3$), high wind speed ($\sim 9.5 \text{ m/s}$) — cities with good natural dispersion
- **Cluster 1 (Moderate Pollution):** Mid AQI (~ 165), average pollutant levels, average wind ($\sim 7.8 \text{ m/s}$) — the most common pollution regime
- **Cluster 2 (Heavily Polluted):** High AQI (~ 245), very high PM_{2.5} ($\sim 194 \mu\text{g}/\text{m}^3$), low wind speed ($\sim 5.9 \text{ m/s}$) — stagnant air trapping pollutants

66. What did PCA help you understand?

PCA reduced the 10 numerical features down to 2 principal components (PC1 + PC2), explaining 21.4% of total variance. The 2D scatter plot showed that the three K-Means clusters are partially separable; Cluster 0 (clean) tends to sit on the left side of the PC1 axis while Cluster 2 (polluted) clusters on the right, with Cluster 1 occupying the middle. This confirmed that the clustering captured real structure in the data. PCA also highlighted that no single pair of dimensions captures the full picture, meaning pollution patterns are driven by a combination of multiple features rather than just one or two.

67. What are the limitations of your analysis?

- Low classifier accuracy
- Arbitrary AQI categories
- Limited PCA variance
- No temporal modeling
- Dataset scope
- Feature independence assumption
- No hyperparameter tuning

Final Submission Details

- **Student Name:** Abdul Rafay
- **Registration Number:** 2280101
- **GitHub Repository Link:** <https://github.com/Seower/AQI-Data-Science-Project>
- **Dataset Used:** Global Urban Air Quality Index Dataset (2015–2025)
- **Best KNN Accuracy:** 0.25
- **Naive Bayes Accuracy:** 0.3866
- **K-Means Clusters Used:** 3
- **PCA Variance Explained:** 21.4%
- **Report File Name:** AQI_Data_Science_Report.pdf